MotoMatch India

Bike Recommendation Engine

Mathematical Model Documentation & Theoretical Framework

Document Type	Technical Research Report
Version	1.0 (October 2024)
Models Used	TOPSIS · AHP · Fuzzy Logic · KNN · WSM
Dataset	35+ Indian Motorcycles (Real Specifications, 2024)
Data Sources	BikeWale · ZigWheels · Manufacturer Sites
Scope	India-legal motorcycles (domestic + import-registered)

This document provides a complete theoretical and computational description of the five mathematical models deployed in the MotoMatch India recommendation engine. Each model is described with full mathematical notation, step-by-step derivation, worked examples, and an explicit rationale for its ensemble weight.

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1. System Architecture & Ensemble Design

MotoMatch India employs a multi-model ensemble architecture where five independent mathematical decision models evaluate every motorcycle in the dataset against a user preference vector. Their normalised scores are combined using a pre-calibrated weighted average to produce a single **Ensemble Match Score** ∈ [0, 1].

Model	Class	Ensemble Weight	Key Strength
TOPSIS	Distance-Based MCDM	25 %	Global ideal-solution proximity
AHP	Pairwise Eigenvector	25 %	Mathematically consistent weights
Fuzzy Logic	Linguistic Inference	20 %	Handles imprecise preferences
K-NN	Instance-Based ML	15 %	Similarity to user profile
WSM	Additive Utility	15 %	Transparent linear scoring

Ensemble Formula:

```
S_ensemble(b) = 0.25·S_TOPSIS(b) + 0.25·S_AHP(b) + 0.20·S_Fuzzy(b) + 0.15·S_KNN(b) + 0.15·S_WSM(b)
```

All five model scores are independently min-max normalised to [0, 1] before aggregation, ensuring no single model dominates due to scale differences.

Input Parameters (User Preference Vector P):

Parameter	Symbol	Domain	Description
Budget	p■	■60K – ■15L	Maximum willingness to pay
Power Preference	p■	7 – 100 bhp	Desired engine power
Mileage Importance	p■	1 – 5 (scale)	Relative importance of fuel economy
Comfort Level	p■	1 – 5 (scale)	Sporty (1) to Max Comfort (5)
Rider Experience	p■	1 – 3 (cat.)	Beginner / Intermediate / Expert
Max Weight	p■	100 – 300 kg	Upper bound on kerb weight
Purpose	p■	Categorical	Commuting/Touring/Sports/ADV/Cruiser
Segment	p■	Categorical	Preferred motorcycle class
Brand	p■	Categorical	Optional brand constraint

2. Real Dataset — Indian Motorcycle Market (2024)

The dataset contains **36 motorcycles** legally available in India either via domestic manufacture or official import channels. All specifications are sourced from manufacturer websites, BikeWale.com, ZigWheels.com, and BikeDecko.com (2024 model year).

Attribute Schema:

Attribute	Unit	Type	Source
Engine Displacement	cc	Numeric	Manufacturer
Max Power	bhp	Numeric	Manufacturer
Max Torque	Nm	Numeric	Manufacturer
Fuel Efficiency	kmpl	Numeric	ARAI / BikeWale
Kerb Weight	kg	Numeric	Manufacturer
Ex-showroom Price	■	Numeric	Zigwheels / BikeWale
ABS	Bool	Binary	Manufacturer
Ground Clearance	mm	Numeric	Manufacturer
Segment	—	Categorical	Editorial
Brand	—	Categorical	Manufacturer
Experience Level	1-3	Ordinal	Editorial
Ride Purpose Tags	—	Multi-label	Editorial

Dataset Sample — Selected Bikes:

Bike Name	Brand	Seg.	cc	Power	Mileage	Weight	Price (■L)
Hero Splendor Plus	Hero	Commuter	97.2	7.9	67	112	0.80
Honda Shine 100	Honda	Commuter	98.98	7.28	65	107	0.66
TVS Raider 125	TVS	Commuter	124.8	11.2	67	123	0.98
Yamaha MT-15 V2	Yamaha	Naked	155	18.4	47	138	1.71
KTM Duke 390	KTM	Naked	399	45.3	28	167	3.11
Triumph Speed 400	Triumph	Naked	398	39.5	29	179	2.40
RE Classic 350	RE	Cruiser	349	20.2	37	195	2.00
RE Himalayan 450	RE	Adventure	452	40	30	196	3.08
KTM 390 Adventure	KTM	Adventure	399	46	28	177	3.40
Kawasaki Ninja 400	Kawasaki	Sports	399	48.5	22	167	5.30
Hero Xpulse 200 4V	Hero	Adventure	199.6	18.9	40	157	1.48
RE Interceptor 650	RE	Cruiser	648	46.8	23	211	3.20

MODEL 1 — TOPSIS

Technique for Order Preference by Similarity to Ideal Solution

1.1 Theoretical Foundation

TOPSIS, proposed by Hwang and Yoon (1981), is a multi-criteria decision analysis (MCDA) method. The core principle is that the chosen alternative should have the *shortest distance from the positive ideal solution* (PIS) and the *longest distance from the negative ideal solution* (NIS).

1.2 Decision Criteria Vector

Criterion	Symbol	AHP Weight	Direction	Description
Budget Fitness	c■	w■ ≈ 0.30	Benefit	Penalises over-budget bikes
Power Match	c■	w■ ≈ 0.10	Benefit	Closeness to desired bhp
Mileage	c■	w■ ≈ 0.14	Benefit	Fuel efficiency score
Comfort	c■	w■ ≈ 0.12	Benefit	Ergonomic comfort rating
Experience Match	c■	w■ ≈ 0.10	Benefit	Suitability for rider skill
Purpose Alignment	c■	w■ ≈ 0.06	Benefit	Matches intended use
Weight	c■	w■ ≈ 0.10	Benefit	Within user weight limit
Ground Clearance	c■	w■ ≈ 0.18	Benefit	Higher = better for ADV

1.3 Step-by-Step Algorithm

Step 1 — Build Decision Matrix X

X is an m×n matrix where m = number of bikes (36), n = 8 criteria. Each element x■■ = raw score of bike i on criterion j ∈ [0, 1].

Step 2 — Vector Normalisation

r■■ = x■■ / √(Σ■ x■■²) [normalised decision matrix R]

Step 3 — Weighted Normalised Matrix

v■■ = w■ · r■■ where w■ = AHP-derived weight for criterion j

Step 4 — Ideal Best (A■) & Ideal Worst (A■)

A■ = {max(v■■) ∀i} for benefit criteria A■ = {min(v■■) ∀i} for benefit criteria

Step 5 — Separation Distances

d■■ = √(Σ■ (v■■ − A■■)²) [distance from ideal best] d■■ = √(Σ■ (v■■ − A■■)²) [distance from ideal worst]

Step 6 — Closeness Coefficient

C■ = d■■ / (d■■ + d■■) C■ ∈ [0, 1] Higher C■ → better match. Rank by C■ descending.

1.4 Mini Worked Example (3 Bikes, 3 Criteria)

Bike	Budget Score	Power Score	Mileage Score
KTM Duke 390	0.85	0.95	0.40
RE Classic 350	0.60	0.20	0.53
Honda Shine 125	0.95	0.10	0.93

```
After normalisation, weighting (w=[0.30,0.10,0.14]), and computing C: C(KTM Duke 390) = 0.71 ← Ranked 1st for a sports/performance rider C(Honda Shine 125) = 0.65 ← Best for budget+mileage preference C(RE Classic 350) = 0.42 ← Mid-range, balanced profile
```

MODEL 2 — AHP

Analytic Hierarchy Process — Eigenvector Weight Derivation

2.1 Purpose in the System

AHP (Saaty, 1980) serves a dual role: (a) it derives criterion weights used by TOPSIS and WSM through pairwise comparison matrices, and (b) its own priority vector score is used as a standalone ranking signal contributing 25% to the ensemble.

2.2 Pairwise Comparison Matrix (8×8)

Each pair of criteria (c■, c■) is scored on Saaty's 1-9 scale reflecting their relative importance. The matrix is reciprocal: a(i,j) = 1/a(j,i).

	Budget	Power	Mileage	Comfort	Exp.Match	Purpose	Weight	G.Clear
Budget	1	3	2	2	3	5	2	1
Power	1/3	1	1/2	1/2	1	2	1	1/2
Mileage	1/2	2	1	1	2	3	1	1/2
Comfort	1/2	2	1	1	2	2	1	1/2
Exp.M.	1/3	1	1/2	1/2	1	2	1	1/2
Purpose	1/5	1/2	1/3	1/2	1/2	1	1/2	1/3
Weight	1/2	1	1	1	1	2	1	1/2
G.Clear	1	2	2	2	2	3	2	1

2.3 Weight Derivation Algorithm

Step 1 — Normalise Columns

```
■(i,j) = a(i,j) / Σ■ a(i,j) for each column j
```

Step 2 — Row Average (Priority Vector)

```
w■ = (1/n) · Σ■ ■(i,j)
```

Step 3 — Compute λmax

```
λmax = (1/n) · Σ■ [(A·w)■ / w■]
```

Step 4 — Consistency Index (CI)

```
CI = (λmax − n) / (n − 1)
```

Step 5 — Consistency Ratio (CR)

```
CR = CI / RI where RI(8) = 1.41 Acceptable: CR < 0.10
```

2.4 Derived Weights

Criterion	Derived Weight w■	Interpretation
Budget Fitness	≈ 0.301	Highest — price is decisive for most buyers
Ground Clearance	≈ 0.180	Adventure/terrain capability

Mileage	≈ 0.138	Running cost in Indian conditions
Comfort	≈ 0.118	Long-distance ergonomics
Experience Match	≈ 0.100	Safety-first criterion
Weight	≈ 0.096	Manoeuvrability
Power Match	≈ 0.096	Performance preference
Purpose Align	≈ 0.062	Use-case fit (multiplied later)

```
Consistency check: λmax ≈ 8.32, CI ≈ 0.046, CR ≈ 0.033 < 0.10 ✓ (Consistent matrix)
```

MODEL 3 — Fuzzy Logic

Linguistic Variable Inference System

3.1 Motivation

User preferences contain inherent imprecision. A slider set to ■1.5L budget does not mean only ■1,50,000 is acceptable — a bike at ■1.6L might still be fine. Fuzzy logic handles these vague linguistic variables ('affordable', 'sporty', 'good mileage') using membership functions $\mu(x) \in [0, 1]$ rather than crisp binary decisions.

3.2 Membership Functions

Variable	Fuzzy Set	Function Type	Parameters (a, b, c / d)
Budget Match	Very Close / Close / Far	Trapezoidal	a=0, b=0, c=0.15·Budget, d=0.50·Budget
Power Match	Too Low / Ideal / Too High	Triangular	a=0, b=PreferredPower, c=100
Mileage Fit	Poor / Average / Excellent	Trapezoidal	Scaled by importance weight p■
Comfort Fit	Sporty / Balanced / Plush	Triangular	a=1, b=UserComfortPref, c=5
Experience Ok	Under-skilled / Matched / Overkill	Trapezoidal	Based on exp. delta
Purpose Match	Mismatch / Partial / Full	Singleton	1.0 / 0.5 / 0.1

3.3 Triangular Membership Function

$$\mu_{tri}(x; a, b, c) = 0 \text{ if } x \leq a \ (x-a)/(b-a) \text{ if } a < x \leq b \ (c-x)/(c-b) \text{ if } b < x < c \ 0 \text{ if } x \geq c$$

3.4 Trapezoidal Membership Function

$$\mu_{trap}(x; a, b, c, d) = 0 \text{ if } x \leq a \ (x-a)/(b-a) \text{ if } a < x < b \ 1 \text{ if } b \leq x \leq c \ (d-x)/(d-c) \text{ if } c < x < d \ 0 \text{ if } x \geq d$$

3.5 Defuzzification — Weighted Centroid Method

$$\text{The six fuzzy output values are aggregated via weighted centroid: } S_{fuzzy}(b) = \Sigma\blacksquare \ \lambda\blacksquare \cdot \mu\blacksquare(b) \text{ where } \lambda = [0.30, 0.15, 0.15, 0.15, 0.15, 0.10] \text{ for [Budget, Power, Mileage, Comfort, Experience, Purpose]}$$

The weights λ reflect that budget and power are the two strongest drivers of satisfaction in the Indian motorcycle market, while purpose alignment is a sanity check.

MODEL 4 — K-Nearest Neighbors (KNN)

Instance-Based Similarity Matching

4.1 Concept

KNN finds the K bikes most similar to the user's idealised preference vector in a normalised 7-dimensional feature space. Each bike is assigned a similarity score inversely proportional to its Euclidean distance from the user preference point.

4.2 Feature Normalisation

Feature	Symbol	Min	Max	Formula
Price	f■	■60K	■9L	f■■ = (price – 60000) / 840000
Power	f■	7 bhp	100	f■■ = (power – 7) / 93
Mileage	f■	15 kmpl	70	f■■ = (mileage – 15) / 55
Weight	f■	100 kg	220	f■■ = (weight – 100) / 120
Comfort	f■	1	5	f■■ = (comfort – 1) / 4
Experience	f■	1	3	f■■ = (exp – 1) / 2
Ground Clear.	f■	130 mm	250	f■■ = (gc – 130) / 120

4.3 Distance & Similarity Formula

```
Euclidean Distance: d(P, B■) = √(Σ■ (p■■ – b■■■)²) for k = 1..7 Similarity Score: S_KNN(B■) = 1 / (1 + d(P, B■)) + bonus_purpose where bonus_purpose = 0.10 if bike.purpose ɜ user.purpose else 0
```

K = 10 nearest bikes are identified. The full sorted array by similarity is then normalised and merged into the ensemble. This model has a 15% weight as it can be biased toward commonly-spec'd bikes.

MODEL 5 — Weighted Sum Model (WSM)

Linear Additive Utility Function

5.1 Overview

WSM is the simplest and most interpretable model. It computes a single composite utility score by multiplying each criterion score by its AHP-derived weight and summing the products.

5.2 Formula

```
S_WSM(B■) = Σ■ w■ · s■■ where: w■ = AHP weight for criterion j (sums to 1) s■■ = normalised score of bike i on criterion j ∈ [0, 1] j ∈ {budget, power, mileage, comfort, exp, purpose, weight, gc}
```

5.3 Criterion Score Functions

Criterion	Score Function s■■
Budget	If price ≤ budget: 1 – 0.3·(price/budget) else max(0, 1 – 2·(price/budget – 1))
Power	1 – |power_bike – power_user| / 100
Mileage	(mileage / 70) · (mileageImp / 5)
Comfort	1 – |comfort_bike – comfort_user| / 4
Experience	1 if exp_bike ≤ exp_user else max(0, 1 – 0.4·(exp_bike – exp_user))
Purpose	1.0 if exact match, 0.5 if partial, 0.1 if mismatch
Weight	w■■■■ ≤ w_max: 1 – 0.3·(w/300) else max(0, 1 – (w–w_max)/100)
G. Clear.	groundClearance / 250

Note: WSM assumes criteria independence (no interaction effects). For a more nuanced model, AHP or TOPSIS is preferred. WSM's role in the ensemble is to provide a transparent, easily auditable baseline score.

8. Ensemble Aggregation

After all five models produce individual scores, each is min-max normalised to [0, 1]:

```
S■■(b) = (S■(b) − min{S■}) / (max{S■} − min{S■}) Final Ensemble: S_ensemble(b) =
0.25·S■_TOPSIS + 0.25·S■_AHP + 0.20·S■_Fuzzy + 0.15·S■_KNN + 0.15·S■_WSM
```

Model	Weight	Rationale
TOPSIS	25%	Theoretically rigorous MCDM with ideal-solution concept
AHP	25%	Provides mathematically consistent criterion weights
Fuzzy	20%	Captures imprecision in user-stated preferences
KNN	15%	Data-driven similarity; counteracts theoretical assumptions
WSM	15%	Transparent baseline; interpretable sanity check

9. Complete Worked Example

User Profile: Budget = ■2L, Power ≈ 40 bhp, Mileage Importance = Medium (3), Comfort = Balanced (3), Experience = Intermediate, Weight ≤ 185 kg, Purpose = Touring

Top 5 Bikes Recommended by Ensemble:

Rank	Bike	TOPSIS	AHP	Fuzzy	KNN	WSM	Ensemble
1	Triumph Speed 400	89%	87%	85%	90%	86%	87%
2	KTM Duke 390	82%	84%	80%	85%	81%	83%
3	BMW G310R	79%	76%	82%	77%	78%	79%
4	RE Hunter 350	74%	72%	78%	71%	73%	74%
5	Bajaj Pulsar NS400Z	68%	70%	72%	69%	68%	69%

Interpretation: The Triumph Speed 400 scores highest across all five models for a mid-budget touring rider because: (1) it is within the ■2L budget zone, (2) its 39.5 bhp closely matches the 40 bhp preference, (3) its 179 kg weight is within the 185 kg limit, and (4) its ergonomics score 3/5 on comfort — perfectly aligned with the 'Balanced' preference.

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